There are three types of cells in the blood. Erythrocytes (red blood cells), which contains hemoglobin, the oxygen carrying red pigments. Leukocytes (white blood cells), which are responsible for the body's defence mechanism, platelets, the small particles involved in blood clotting. Erythrocytes, the oxygen carrying cells of the blood have the dominating presence. In fact, 99 per cent of the cells are erythrocytes. These cells have high surface- to – volume ratio. This enables them to rapidly diffuse oxygen and carbon dioxide to and from the interior of the cells. The erythrocytes are produced in the soft interior of bones called bone marrow. Erythrocytes are immature cells. These cells lack nucleus and organelles. They cannot reproduce themselves and their life-span is also short. The average life-span of any erythrocyte is approximately 120days. The total volume of circulating erythrocytes in human body remains fairly constant.

The chief characteristic of erythrocytes is the presence of protein, hemoglobin, which binds oxygen taken in by the lungs. In a hemoglobin molecule, there are four sub-units. Each of the four sub-units consists of an organic molecule known as heme attached to a polypeptide. The four identical polypeptides join together to form an entity known as globin. Each heme portion contains an iron atom. It is the iron atom which binds oxygen. The cytoplasmic granules of the neutrophils are packets of potent enzymes capable of digesting many types of cellular materials. Like neutrophils, eosinophils are also highly mobile and capable of engulfing and destroying micro-organism. Monocytes are the largest cells of the blood. They have a relatively large nucleus.

SPEED PRACTICE – 4:

SPEED PRACTICE - 4 Time - 15 Minutes

Type the following in double line spacing with a margin of 10 degrees:-

Gibbs was born on February 11, 1839 in New Haven, U.S. of a distinguished and learned family. His father was a professor in the school of Divinity at Yale and a linguist. The young Gibbs excelled in classics and graduated in 1858, winning prizes in mathematics and Latin, from the Yale College. He displayed an uncommon talent for geometry which is professor Hubert Newton encouraged. Since the physics curriculum at Yale was so elementary at that time, he worked for and earned his doctorate in; engineering (1863). Thereafter, he was tutor in the college teaching Latin for two years and natural philosophy for a year.

From 1866 to 1869 he studied at the Universities of Berlin, Heidelberg and Paris, where his teachers were some of the world's eminent mathematicians and physicists. In 1871, Yale offered him the new chair of mathematical physics, the post provided no salary. Yet Gibbs, who had independent means and an umbilical attachment to New Haven, accepted the post. The college complimented him with a salary of 2000 dollars only after ten years of service. Except for the period1866 to 1869 he spent in Europe, he served as Professor of mathematical physics in Yale College for the rest of his life till April 28, 1903. With his training as an engineer, Gibbs took up the development of a governor for steam engines, which launched him into the field of thermodynamics. To express the basic

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